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Statement on Mentoring a Diverse Student Body

Students arrive at college with different levels of preparation and familiarity with academic culture. These differences often reflect the opportunities they have or have not had, rather than their potential. I believe it is my responsibility as an educator to reduce the extent to which prior access determines who can thrive in computing. I approach this responsibility through two complementary goals: helping students develop a mastery orientation toward learning, and encouraging adaptive help-seeking.

In my teaching, I emphasize developing understanding rather than demonstrating ability. I design learning objectives and assessments that help students recognize their progress and identify what they have yet to understand. My dialogue–practice–reflection approach gives students repeated opportunities to explain their reasoning and revise it in response to feedback. I adapt the support I provide to different audiences without reducing the intellectual challenge. Holding the challenge constant matters because differences in prior exposure to computing can otherwise be mistaken for differences in aptitude, by instructors and by students themselves.

I see early undergraduate research as a way to build confidence and a sense of belonging. Working alongside faculty allows students to see that experienced researchers also encounter uncertainty and seek guidance. Students should not have to decide they are "research people" before trying research; experiencing it firsthand is often how they find out. I would actively recruit students without prior research experience, particularly those from lower-income and first-generation backgrounds and from groups underrepresented in computing. I would also seek institutional and external funding, including NSF support, to offer paid research positions. Making opportunities visible is insufficient if students cannot afford to participate.

Research mentorship allows me to individualize support, including adapting to how each student works best. Some students I have mentored preferred structured weekly meetings; others organized their work through detailed task lists on a shared board. For students who found it hard to sustain momentum on their own, I offered body-doubling sessions, working alongside them in real time, which helped them make steady progress.

My research seeks to demystify performance engineering by making expert reasoning explicit, and I bring the same commitment to mentoring. Students sometimes arrive having written code and generated results but unsure how to interpret them, or believing that negative results indicate failure. Since 2025, I have used research notebooks with the undergraduate and two master's students I have mentored, as well as with PhD students I support through peer mentoring in my research group. Students record their hypotheses and what they learn from each experiment, which helps them identify gaps in their reasoning and design new experiments independently. One undergraduate I mentored

began by building an automatic code generator. Reviewing their results, they recognized that the generator made it possible to specialize implementations to specific problem sizes and hardware, and identified a question worth investigating: how to balance the performance gains of many specialized implementations against the overhead they add to a codebase. I treat requests for help as opportunities to reason alongside students rather than simply provide answers, so that seeking guidance becomes part of developing independence.

My experience as an international graduate student has shown me how isolation and unspoken norms, such as whether it is acceptable to ask a peer to read a draft, can quietly shape who seeks help. As a faculty member, I would make these norms explicit: celebrating paper submissions as well as acceptances, inviting students to share drafts early, and reaching out deliberately to students who are far from their home communities.

Across teaching and mentoring, my aim is to make visible the reasoning and the unwritten rules that experienced researchers take for granted, so that students can develop the confidence and independence to pursue paths they may not initially have imagined for themselves.